

Design Proposal: Mach–Zehnder Interferometers for Extracting the Group Index of Silicon Strip Waveguides

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Abstract—This proposal describes a baseline unbalanced Mach–Zehnder interferometer (MZI) for extracting the group index n_g of a 500×220 nm silicon strip waveguide (quasi-TE, 1550 nm), and variations of it, for the openEBL-2026-10 run (605×410 μm allocation). The baseline uses Y-branch splitters and a path-length difference $\Delta L = 100$ μm . Lumerical MODE gives $n_g = 4.180$ at 1550 nm, so the baseline free spectral range (FSR) is 5.75 nm. The variations change the path-length difference (0, 50 and 150 μm), the splitter (Dream Photonics 1×2 splitter; broadband directional coupler with $\Delta L = 200$ μm) or the waveguide width (480 nm). The layout passes design-rule and functional verification, and Lumerical INTERCONNECT simulations of the layout give FSRs within 1% of the design values.

Index Terms—Silicon photonics, Mach–Zehnder interferometer, group index, free spectral range, compact model, electron-beam lithography.

I. INTRODUCTION

THE group index n_g sets the free spectral range of interferometers and resonators such as MZIs and ring filters. An unbalanced MZI is the simplest circuit for measuring n_g : its transmission is periodic in wavelength with a period (the FSR) set by n_g and the drawn path-length difference ΔL .

The design is built around one baseline device, DEV1: an MZI with two Y-branches, 500 nm waveguides and $\Delta L = 100$ μm . The other devices are variations of the baseline (Table I), used to

- 1) check the FSR-versus- ΔL trend (DEV2, DEV4) and the insertion loss without fringes (DEV3);
- 2) measure the change of n_g with width (DEV5); and
- 3) compare splitters (DEV6, DEV7).

All structures fit in the 605×410 μm allocation and use SiEPIC EBeam PDK cells [3].

II. THEORY

A. Waveguide

The waveguide is a fully etched silicon strip, 220 nm thick and $w = 500$ nm wide, on a buried oxide and covered by an oxide cladding [1], [2]. The propagation constant is $\beta(\lambda) = 2\pi n_{\text{eff}}(\lambda)/\lambda$. The group index is

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \quad (1)$$

B. MZI transfer function

Consider an MZI with two ideal, lossless 50/50 1×2 splitters (e.g., Y-branches) and arms of length L_1 and $L_2 = L_1 + \Delta L$ with power attenuation α . The output field is

$E_o = \frac{1}{2} E_i (e^{-j\beta_1 L_1 - \alpha_1 L_1/2} + e^{-j\beta_2 L_2 - \alpha_2 L_2/2})$. The transmission is therefore

$$T_{1 \times 2}(\lambda) = \frac{1}{4} \left| e^{-j\beta_1 L_1 - \alpha_1 L_1/2} + e^{-j\beta_2 L_2 - \alpha_2 L_2/2} \right|^2, \quad (2)$$

and when there is negligible loss, i.e. $\alpha_1 = \alpha_2 = 0$, it reduces to

$$T_{1 \times 2}(\lambda) = \frac{1}{2} \left[1 + \cos \left(\frac{2\pi n_{\text{eff}}(\lambda)}{\lambda} \Delta L + \phi_H \right) \right], \quad (3)$$

where ϕ_H is any additional, wavelength-independent phase difference between the arms (e.g., from fabrication) that shifts the fringes but not the FSR.

For an MZI built from two identical, lossless 2×2 directional couplers, each coupler relates the fields at its two outputs to those at its two inputs by

$$\begin{pmatrix} E_{\text{out},1} \\ E_{\text{out},2} \end{pmatrix} = \begin{pmatrix} t & -j\kappa \\ -j\kappa & t \end{pmatrix} \begin{pmatrix} E_{\text{in},1} \\ E_{\text{in},2} \end{pmatrix}. \quad (4)$$

Here t is the real field self-coupling (straight-through) coefficient and κ the real field cross-coupling coefficient, with $t^2 + \kappa^2 = 1$: a fraction κ^2 of the power crosses to the other waveguide, and the crossed field picks up a phase of $-\pi/2$ (the factor $-j$). Light propagating along arm i ($i = 1, 2$, of length L_i) is multiplied by the arm's complex field transmission

$$a_i = e^{-j\beta L_i} e^{-\alpha L_i/2}, \quad (5)$$

whose phase is set by $\beta = 2\pi n_{\text{eff}}/\lambda$ and whose amplitude is reduced by the power attenuation coefficient α (the field decays at $\alpha/2$). Launching light into input 1 and applying (4), then (5), then (4) again gives the bar (output 1) and cross (output 2) transmissions

$$T_{\text{bar}} = |t^2 a_1 - \kappa^2 a_2|^2, \quad T_{\text{cross}} = |t\kappa(a_1 + a_2)|^2. \quad (6)$$

For $\kappa^2 = \frac{1}{2}$ these become $\sin^2(\beta\Delta L/2)$ and $\cos^2(\beta\Delta L/2)$.

C. Group index from the free spectral range

Transmission minima of (3) occur when $\beta(\lambda_m)\Delta L = (2m + 1)\pi$. For two adjacent minima $\lambda_1 < \lambda_2$,

$$\beta(\lambda_1)\Delta L - \beta(\lambda_2)\Delta L = 2\pi. \quad (7)$$

Approximating that β varies linearly with λ about λ_1 gives $\beta(\lambda_2) \approx \beta(\lambda_1) + \frac{d\beta}{d\lambda}(\lambda_2 - \lambda_1)$, and

$$\frac{d\beta}{d\lambda} = \frac{d}{d\lambda} \left(\frac{2\pi n_{\text{eff}}}{\lambda} \right) = -\frac{2\pi}{\lambda^2} \left(n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda} \right) = -\frac{2\pi n_g}{\lambda^2}. \quad (8)$$

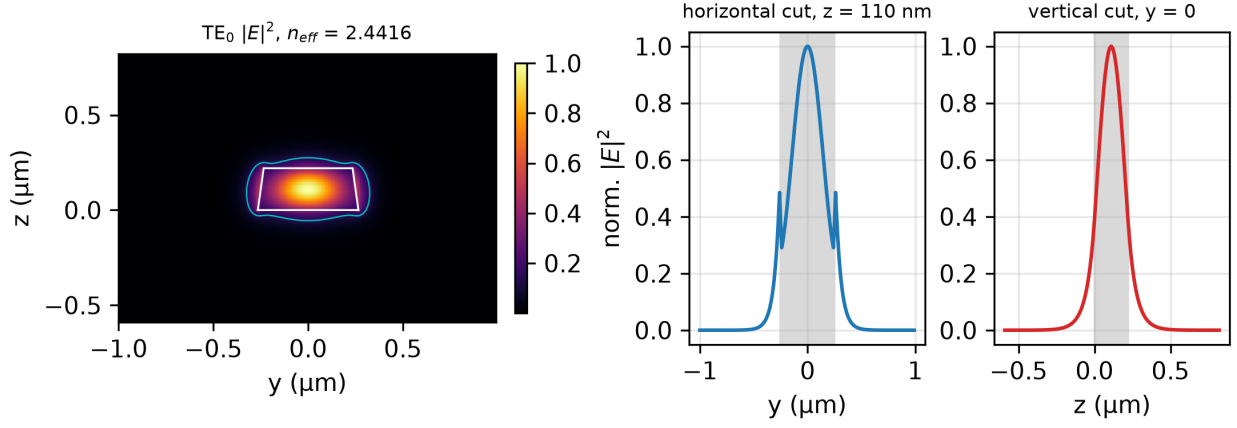


Fig. 1. Fundamental quasi-TE mode of the 500×220 nm strip waveguide at 1550 nm (Lumerical MODE): normalized $|E|^2$ with the $1/e^2$ contour (cyan) and core outline (white), and horizontal and vertical cuts through the mode centre (grey: silicon core).

Substituting into (7) with $\text{FSR} = \lambda_2 - \lambda_1$ gives $2\pi n_g \Delta L \text{FSR} / \lambda^2 = 2\pi$, so

$$\text{FSR} = \frac{\lambda^2}{n_g \Delta L} \iff n_g(\lambda) = \frac{\lambda^2}{\Delta L \cdot \text{FSR}(\lambda)} \quad (9)$$

Here λ is evaluated at the centre of the fringe pair.

The 50 nm measurement bandwidth requires $\text{FSR} < 50$ nm, i.e. $\Delta L > \lambda^2 / (n_g \cdot 50 \text{ nm}) = 11.5 \mu\text{m}$. The designs use FSRs of 2.9–11.5 nm.

III. BASELINE DEVICE (DEV1)

A. Waveguide

The waveguide cross-section was simulated with the Lumerical MODE eigenmode solver, using the course setup scripts and the process geometry of the PDK [8]: a 220 nm-thick Si core with 82° sidewalls (mean width equal to the drawn width; dispersive Lorentz model, $n_{\text{Si}} = 3.4734$ at 1550 nm) on a $3.017 \mu\text{m}$ buried oxide, with $n_{\text{SiO}_2} = 1.444$ below and above. The mesh is 10 nm. The fundamental quasi-TE mode at 1550 nm (Fig. 1) has $n_{\text{eff}} = 2.4416$.

A MODE frequency sweep from 1500 to 1600 nm (Fig. 2), fitted with a second-order polynomial about $\lambda_0 = 1.55 \mu\text{m}$, gives the compact model

$$n_{\text{eff}}(\lambda) = 2.44157 - 1.12155(\lambda - \lambda_0) - 0.03412(\lambda - \lambda_0)^2, \quad (10)$$

with λ in μm . Inserting (10) into (1) gives

$$n_g(\lambda_0) = n_1 - n_2 \lambda_0 = 2.44157 + 1.12155 \times 1.55 = 4.1800. \quad (11)$$

B. Circuit

DEV1 is the MZI of Fig. 3(a): an input grating coupler (GC), a Y-branch (ebeam_y_1550 [5]), two arms of 132.397 and $232.397 \mu\text{m}$ ($\Delta L = 100 \mu\text{m}$), a second Y-branch and an output GC. The arms use the PDK strip waveguide ($5 \mu\text{m}$ -radius Bezier bends), with the same bends in both arms so that ΔL is straight waveguide. With $n_g = 4.180$, (9) gives $\text{FSR} = 5.748$ nm at 1550 nm.

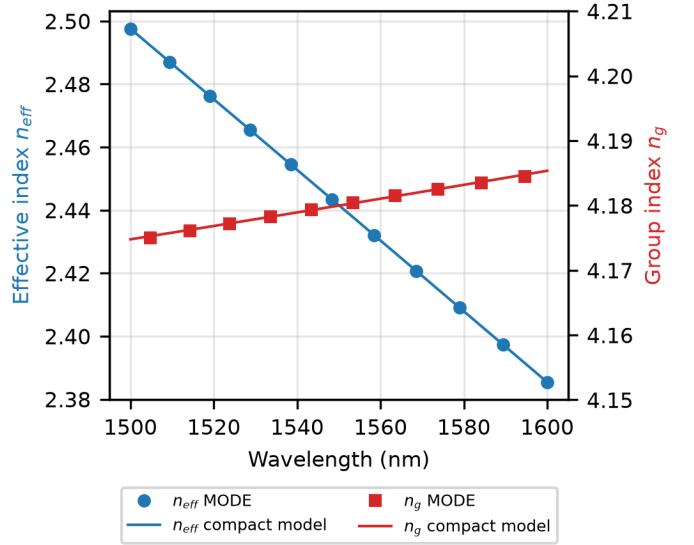


Fig. 2. Effective index (left axis) and group index (right axis) of the 500×220 nm TE mode: Lumerical MODE frequency sweep (markers) and the compact model (10) (lines).

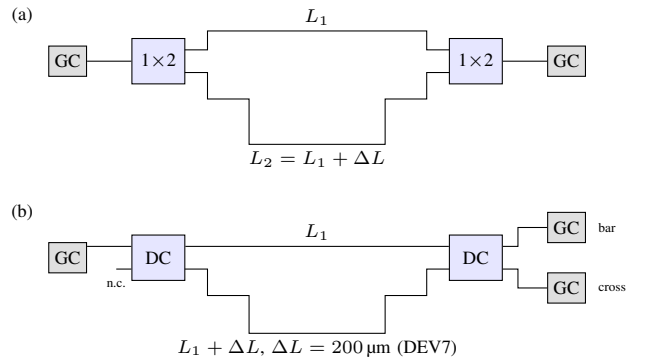


Fig. 3. Schematics (not to scale). (a) Baseline MZI with 1×2 splitters: Y-branches (DEV1–DEV5) or Dream Photonics 1×2 splitters (DEV6). (b) MZI with two 2×2 broadband directional couplers (DEV7).

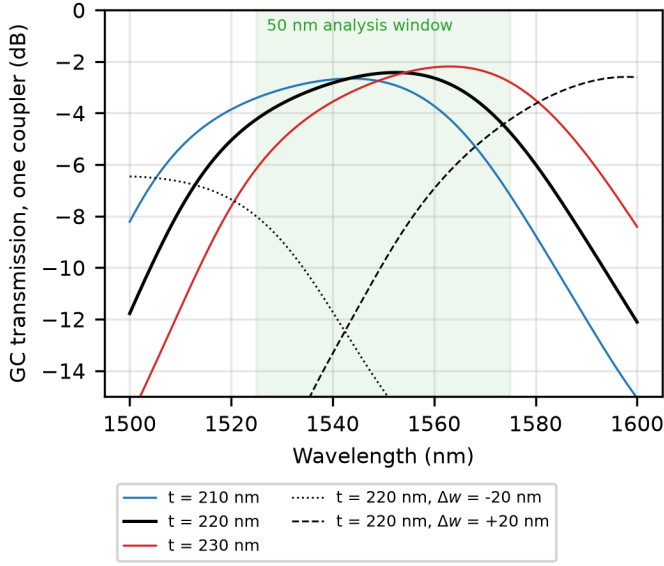


Fig. 4. Transmission of one TE 1550 grating coupler (PDK compact model) for Si thickness 210/220/230 nm and tooth-width bias $\Delta w = \pm 20$ nm.

C. Grating couplers

All circuits use the PDK TE 1550nm grating coupler GC_TE_1550_8degOxide_BB, a black-box cell that the foundry replaces with its own design [4]. The GCs face right on a 127 μ m pitch to match the fibre array. Fig. 4 shows the PDK compact model of one GC (peak loss 2.4dB at 1552.5nm). The analysis window is centred on the peak of the measured loopback.

D. Extracting n_g

The chips will be measured on the automated SiEPIC test setup: a laser sweep from 1499 to 1601nm in 5000 points (20.4pm steps), set by the Phot1x test rules. Each spectrum will be processed as follows:

- 1) convert to dB and remove the GC envelope, by subtracting the loopback (CAL1) or a polynomial fitted through the fringe maxima;
- 2) locate each transmission minimum, refined by a parabolic fit to ± 3 samples of the normalized transmission in linear units;
- 3) calculate $\text{FSR}_i = \lambda_{i+1} - \lambda_i$ and n_g from (9) at $(\lambda_{i+1} + \lambda_i)/2$.

IV. VARIATIONS

Table I lists the baseline and its variations; Fig. 6 shows the FSR of each ΔL .

Path-length difference (DEV2, DEV3, DEV4). DEV3 ($\Delta L = 0$) gives the insertion loss of the circuit without fringes. DEV4 ($\Delta L = 50\mu\text{m}$, FSR 11.5nm) and DEV2 ($\Delta L = 150\mu\text{m}$, FSR 3.83nm) give the FSR-versus- ΔL trend with the baseline. The long arm of DEV2 is folded upward (Fig. 8), with four bends in each arm.

Width (DEV5). DEV5 draws both arms at 480nm, with identical 10 μ m tapers at the splitter ports. MODE gives $n_g = 4.215$ at 480 nm (Fig. 7).

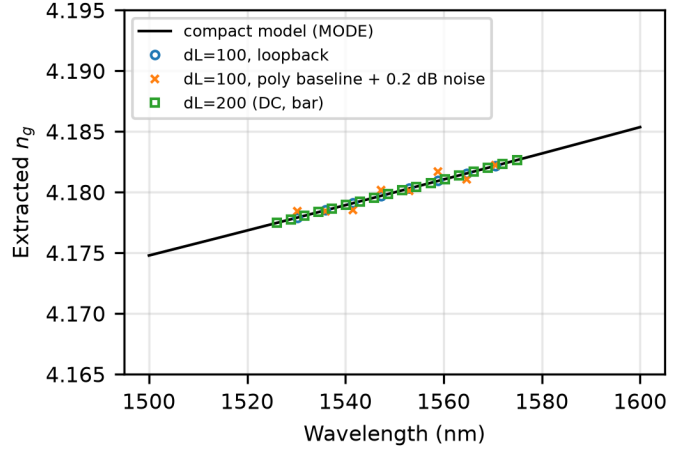


Fig. 5. Group index extracted with (9) from simulated spectra on the measurement grid, compared with the compact model.

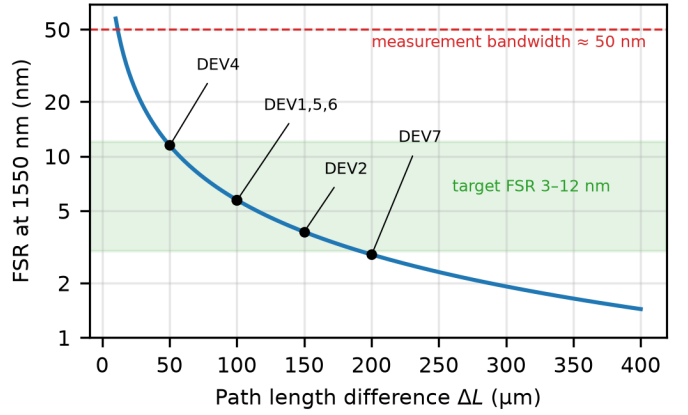


Fig. 6. FSR at 1550 nm versus path-length difference for the 500 nm waveguide ($n_g = 4.180$). Dots: designs.

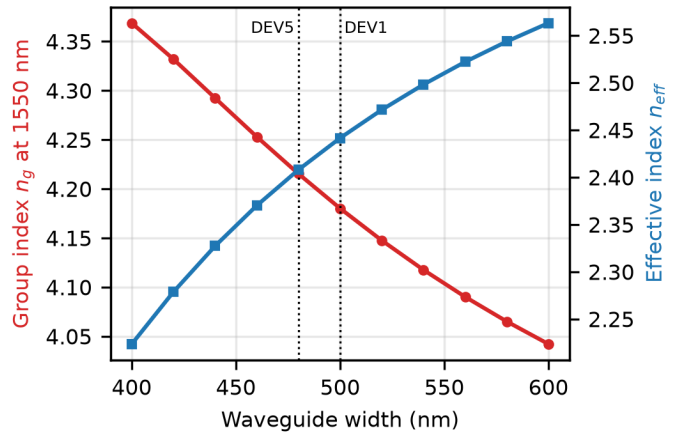


Fig. 7. Group index (left) and effective index (right) of the TE mode at 1550 nm versus strip width (Lumerical MODE, 220 nm thick).

Splitter (DEV6, DEV7). DEV6 replaces the Y-branches with the Dream Photonics 1×2 splitter (ebeam_dream_splitter_1x2_te1550_BB), an allowed black-box cell of the run [4]. DEV7 uses two broadband directional couplers (ebeam_bdc_te1550 [6], Fig. 3(b)), with

TABLE I
BASELINE DEVICE AND VARIATIONS (TE, 1550 nm, 220 nm Si)

Device	Change from baseline	ΔL (μm)	Width (nm)	n_g (sim.)	FSR (nm)	Purpose
DEV1	— (baseline: Y-branches)	100	500	4.180	5.748	Primary n_g extraction
DEV2	$\Delta L = 150 \mu\text{m}$	150	500	4.180	3.832	FSR-versus- ΔL trend
DEV3	$\Delta L = 0$ (balanced)	0	500	4.180	—	Insertion-loss reference
DEV4	$\Delta L = 50 \mu\text{m}$	50	500	4.180	11.50	FSR-versus- ΔL trend
DEV5	width 480 nm	100	480	4.215	5.700	n_g versus width
DEV6	Dream 1×2 splitters	100	500	4.180	5.748	Splitter comparison
DEV7	broadband DCs, $\Delta L = 200 \mu\text{m}$	200	500	4.180	2.874	Splitter comparison
CAL1/CAL2	GC loopbacks (no MZI)	—	500	—	—	GC envelope and repeatability

TABLE II
PROCESS CONSTRAINTS AND DESIGN

Constraint	Source	Design
Si thickness 220 nm, $6\sigma = 22.3$ nm	[8]	MODE with 220 nm; $\pm 1\sigma$ (3.7 nm) gives $n_g = 4.1772 / 4.1842$
Sidewall angle 82° , single full etch	[8]	MODE core is a trapezoid with 82° sidewalls
Buried oxide $3.017 \mu\text{m}$; oxide cladding	[8]	Included in the MODE geometry
Minimum feature 60 nm	[8]	DRC: 0 errors
Shot pitch 6 nm; periodic structures in multiples of it	[8]	No periodic structures drawn; DEV5 width 480 nm = 80×6 nm; 500 nm waveguides (DEV1–DEV4, DEV6, DEV7) snap to 498 or 504 nm, giving $n_g = 4.173$ – 4.184 ; SEM requests on DEV1 and DEV5 measure the fabricated widths
Measured $n_g = 3.17$ – 4.26 (TE, 1550 nm)	[9]	FSRs 2.8–15.2 nm over this range, all below 50 nm
Y-branch loss 0.11–0.31 dB; waveguide loss 0.71–2.89 dB/cm	[9]	At most 0.62 dB (two Y-branches) and 0.12 dB (longest arm, 412 μm) per MZI
GCs facing right, 127 μm pitch, fibre 2 input, opt_in label format	[4]	All GCs face right; input GC on fibre 2 with outputs on fibre 1 (DEV7: fibres 3–4); functional verification: 0 errors

both outputs routed to GCs and the unused input terminated.

Calibration (CAL1, CAL2). The two GC loopbacks give the GC spectral envelope and the GC-to-GC repeatability. The measured GC peak wavelength (simulated 1552.5 nm, Fig. 4), the n_g of DEV1 and the SEM width of DEV1 will be used together to estimate the fabricated Si thickness and width.

V. LAYOUT AND FABRICATION

A. Process constraints

Table II lists the constraints of the PDK document [8], the process-control data of past runs [9] and the test rules of the run [4], and how the design meets each.

B. Layout

The layout (Fig. 8) was generated with SiEPIC-Tools [7] and the SiEPIC EBeam PDK. GCs sit in columns of four slots at 127 μm pitch, and each device has its input GC directly below its output GC (DEV7: two output GCs below its input GC). Each input GC carries a test label, e.g. opt_in_TE_1550_device_JeremyLeow_DEV1dL100. All drawn ΔL equal the design values to 1 nm. Two SEM requests (layer 200/0) are placed on the arms of DEV1 and DEV5 to measure the fabricated widths.

C. Verification

SiEPIC functional verification with the Phot1x test rules (58 components, 9 test labels) and a DRC with the PDK's Si rules (minimum width 60 nm, minimum space 70 nm, all Si inside the floorplan) both report 0 errors. The openEBL-2026-10 submission scripts [4] also report 0 errors, with all 21 black-box cells replaceable.

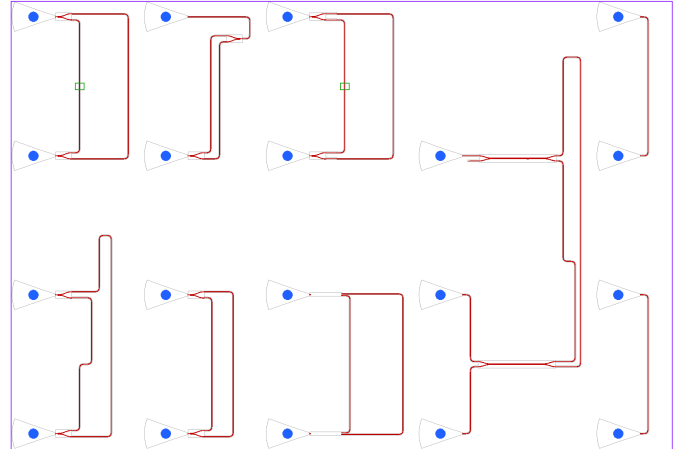


Fig. 8. Layout of the $605 \times 410 \mu\text{m}$ cell: Si (red), device outlines (grey), GC fibre targets (blue) and SEM requests (green). Columns from left (upper / lower): DEV1 / DEV2; DEV3 / DEV4; DEV5 / DEV6; DEV7; CAL1 / CAL2.

D. Simulation of the layout

The netlist of each circuit was extracted from the layout with SiEPIC-Tools and simulated in Lumerical INTERCONNECT with the PDK compact model library, sweeping 1499–1601 nm in 5000 points. Fig. 9 shows the setup for the baseline, Fig. 10 the spectra and Table III the results.

Table III lists the FSR, the n_g extracted with (9) and the peak insertion loss of each circuit.

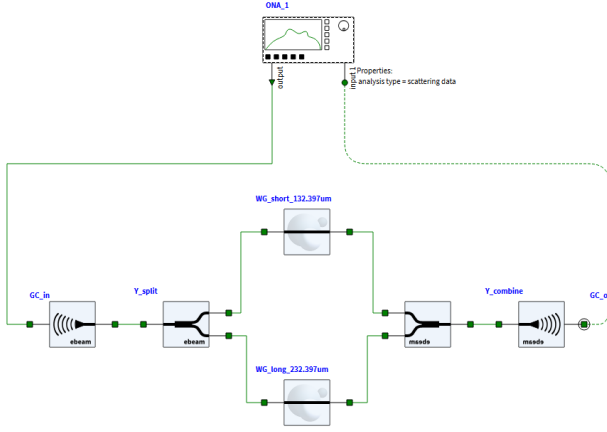


Fig. 9. Lumerical INTERCONNECT setup of the baseline DEV1: Optical Network Analyzer (ONA_1), grating couplers, Y-branches, and the two arm waveguides labelled with their drawn lengths.

TABLE III
INTERCONNECT SIMULATION OF THE LAYOUT

Device	ΔL (μm)	FSR (nm)	n_g	Peak IL (dB)
DEV1	100	5.733	4.1904	5.16
DEV2	150	3.821	4.1921	5.12
DEV3	0	—	—	5.10
DEV4	50	11.462	4.1921	5.20
DEV5	100	5.682	4.2286	5.31
DEV6	100	5.733	4.1904	4.85
DEV7	200	2.849	4.2169	5.03
CAL1/CAL2	—	—	—	4.77

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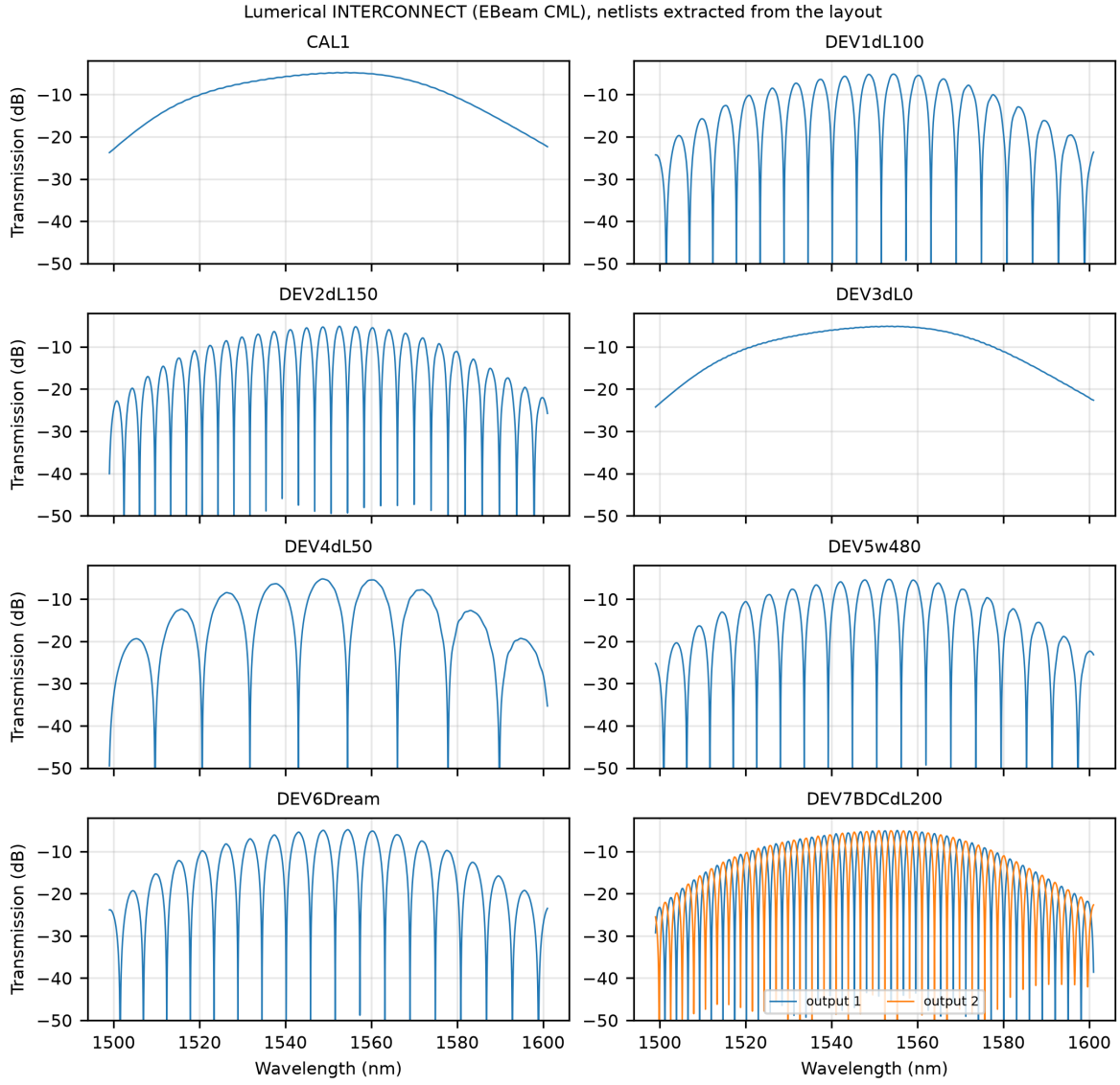


Fig. 10. Lumerical INTERCONNECT transmission spectra of every circuit, simulated from the netlists extracted from the layout (including the GC response).